

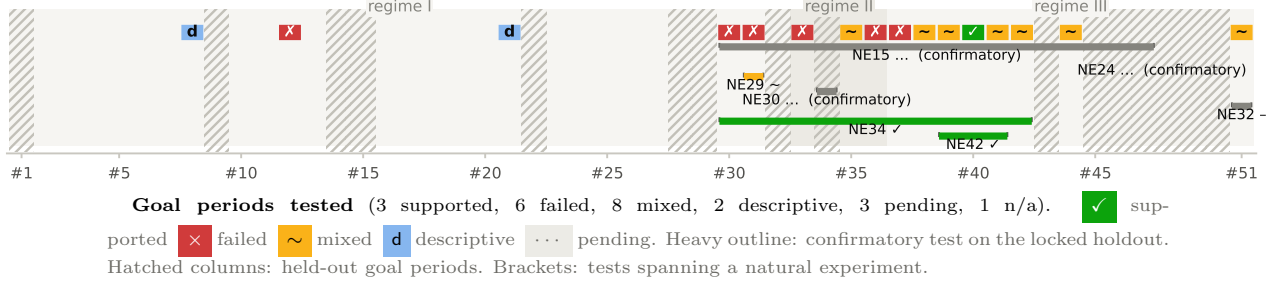
H01: Emergent superagents exist

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 2/5

Bottom line. No group is an effective superagent: one agent plus its own artifact persists, the goal prompt drives it, and grouping adds no measurable agency.

level: hypothesis · pipeline checks 84% · updated 2026-10-04



WHAT HAPPENED

Round 2 asks whether a group of agents plus its shared artifacts forms an *effective superagent*. Such a group uses Kolchinsky–Wolpert (KW) semantic information to stay viable and holds that information at the group level. The data is the work ledger: 58k strict git/deploy writes in 19 non-holdout units (#30–#51). The candidates are artifact crews and behavior-synchrony, co-allocation and reply communities. Rooms and labs are baselines. **No candidate is an individuality maximum:** crews give Stouffer $Z = +0.06$, and 2–8% are local maxima. No candidate carries KW semantic information about its environment ($\Delta V_{st} > 0$ in 2/19 units, $\Delta V_{obs} > 0$ in 0/19). No candidate repairs itself beyond aggregation. Allocation persists across nights ($\Delta C = 0.47$) and memory loss (16/17), but single agents out-persist crews. A goal change cuts advancement by 0.39. The holdout tests (NE30, NE24) exist and are **not run**.

PHYSICAL MODEL

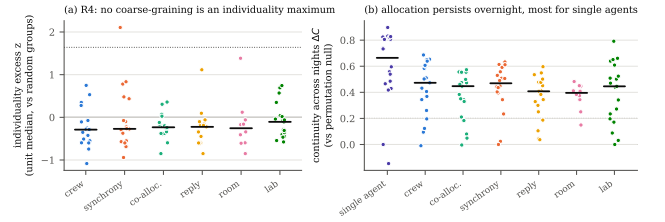
Model 04 (KW semantic information). The system is a coarse-grained group G : members, shared artifacts R_G and channel. The state x_b is binary: G advances R_G in 30-min bin b , or not. The environment y_b is the rest of the swarm’s writes plus human and automated messages. The viability V is advancement over horizon τ . p^f is KW’s coarse-graining scramble of $p(x_0, y_0)$. A is Krakauer et al.’s colonial individuality: self-prediction beyond the environment. The share $\iota = A/H(x')$ and its z -scores are project choices, not the paper’s. ΔC is the return to R_G after a night, above a permutation null.

$$\Delta V_{st} = V[p] - V[p_X \otimes p_Y] = \mathbb{E}_p[(1 - e^{-\text{PMI}}) v(x_0, y_0)], \quad |\Delta V| \leq \sqrt{I/2}$$

$$S = \min_f \{I(p^f) : V[p^f] \approx V[p]\}, \quad \eta = S/I(X_0; Y_0)$$

$$\Delta V_{obs} : p(x_{t+1}|x_t, y_t) \rightarrow p(x_{t+1}|x_t)$$

$$A_G = I(x_{b+1}; x_b | y_b), \quad \iota_G = A_G/H(x_{b+1})$$



Observables. (a) Each dot is the median individuality excess of one unit’s groups over activity-matched random groups. Every coarse-graining sits at the null. (b) The return to the group’s artifacts the next morning, above a permutation null. Membership excludes the scored day. Allocation persists in every unit, and most for single agents with their own projects. Rooms and labs are not special.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

No group-level mind exists to steer. Coordination on artifacts adds no measurable self-maintenance beyond the members. Each agent returns to its own artifacts after every night’s context erasure ($\Delta C = 0.47$) and after memory loss (16/17). The operator’s goal sets the work rate: a goal change cuts advancement by 0.39. To keep work going across resets, write the work state into the repo. Do not rely on memory or on team culture. The KW values are near zero partly because the ledger counts attempts, not outcomes.

RESULTS VS PREDICTIONS

prediction	observed	verdict
R4 crews are max- ima	$Z = +0.06$; first in 5/18; 3.5% local max	failed [†]
R4d artifact groups persist most	4/19; single agents top in 15/18	failed
R5a/b $\Delta V_{st}, \Delta V_{obs} > 0$	2/19 and 0/19	failed
R5c goal change costs viability	-0.39 ± 0.14 , $z = -2.75$	supported
R6b crews persist overnight	$\Delta C \geq 0.2$ in 15/19 (median 0.47)	supported
R6c survives mem- ory loss	16/17 vs base 0.91	weak supp.
R6a/R8 spillover, repair	$Z = 2.4 < 3$; no repair	failed
R7 substrate inde- pendence	departures free; no turnover	mixed

[†]validated power $\leq 7\%$ at village sampling.

GOAL PERIODS IN DETAIL

NE15 ... pending (**holdout**)
 NE24 ... pending (confirmatory, not run) (**holdout**)
 NE30 ... pending (confirmatory, not run) (**holdout**)
 G08 d 1b: descriptive (unchanged; P4 still not as predicted)
 G12 × 1b: failed (no team unit beyond the re-partition null...
 G21 d 1b: descriptive (unchanged)
 G30 × failed (round 2: no agency signature)
 G31 × failed (round 2: no agency signature)
 G33 × failed (round 2: no agency signature)
 G35 ~ 1b: mixed (unchanged, both models)
+13 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

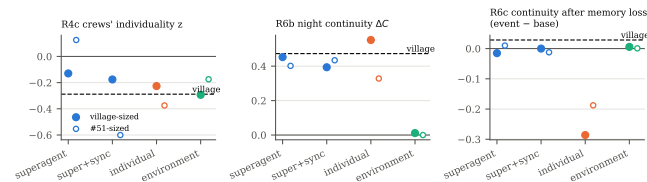
No replay exists. A goal change bundles a new instruction with the scramble. The only artifact-store scramble (NE24) is in the holdout. The KW values rest on a coarse Markov kernel, and its bootstrap is anti-conservative under hidden state. The binary state cannot see *which* artifact a group works on. The environment *y* excludes the scheduler, kickoff and prior fields, so colonial *A* can still contain field-driven persistence. The ledger counts attempts: it has no failures, no value and no non-git work. The memory-loss evidence is 17 events, mostly in #51, where private goals can carry continuity. **Round 1b** repeats the round-1 content tests on improved data with two embedding models. Rooms stay ordered content groups after style removal. Lab order is style. The drafted #12 teams are not groups. The NE42 merge-split shows that room order follows the channel. Part of the round-1 merge DiD is style.

ROUND 2

Run `confirm_r2.py`. H58 runs a coordination-first search with a which-artifact state. Recompute KW with the outcome and context ledgers.

- **Essence.** effective superagents in Kolchinsky–Wolpert terms, running on a substrate of agents.
- **H01-R4** Individuality maxima pick out candidate superagents.
- **H01-R5** Superagents have positive group-level semantic information.
- **H01-R6** The agency is effective: it lives above the substrate.
- **H01-R7** Substrate independence.
- **H01-R8** The superagent does self-maintenance.

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Second figure. Synthetic worlds at village sampling (filled) and #51 sampling (open). The dashed line is the village. Timing individuality does not separate the worlds. Night continuity rules out the purely environment-driven world. Continuity after memory loss rules out memory-held plans. The village sits with the artifact-held worlds.